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The chemistry of "the Mother of Satan": A comprehensive review on triacetone triperoxide (TATP)

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ABSTRACT

This review provides a comprehensive and critical overview of the chemistry of **TATP** – known as "the Mother of Satan" - integrating historical context, synthetic pathways, conformational and polymorphic behavior, physicochemical properties, detonation mechanisms, and deactivation strategies. Particular emphasis is placed on the stereoelectronic origins of instability, entropy-driven detonation behavior, phase and gas-phase phenomena preceding initiation, and the narrow boundary between controlled decomposition and accidental detonation. Advances in analytical methodologies for detection and quantification are discussed, highlighting the intrinsic challenges posed by volatility, lack of chromophores, and rapid degradation. Strategies for chemical and materials-assisted mitigation, process safety, and emergency response are also examined. By consolidating experimental, theoretical, and forensic insights, this review aims to provide a unified chemical framework for understanding **TATP** behavior and to support informed risk assessment, detection, and countermeasure development in both laboratory and security contexts.

1. Initial considerations

Terrorism remains a persistent and complex global challenge, affecting not only the safety of populations but also the institutional, political, and economic structures of many countries. The consequences of terrorist attacks are not limited to the physical destruction they cause, since they also generate profound indirect effects, altering individual behavior, reshaping political attitudes, and redirecting national priorities [1].

Over the past two decades, indirect consequences of terrorism have been empirically associated with increased outgroup hostility and xenophobia, rise of political conservatism, and rally-'round-the-flag effects in Western and other societies, especially in response to Islamist violence [2]. These shifts in public sentiment have influenced electoral outcomes, reinforced authoritarian discourse, and prompted the implementation of restrictive security and migration policies [3].

Beyond these sociopolitical effects, terrorism disrupts innovation and economic development. It redirects public resources away from long-term investment areas, such as in green innovation, induces economy-wide uncertainty, and restrains international flow of capital and knowledge, ultimately inhibiting sustainable growth [1]. Countries with high-tech manufacturing capacity and export sectors have been shown to suffer particularly from these disruptions, although poor and underdeveloped countries have not been spared as well.

Terrorism is generally defined as the premeditated use or threat of violence against noncombatants to obtain political or social objectives through the intimidation of a broader audience [3]. This distinguishes it from other forms of violence, such as organized crime or civil conflict. Since the 9/11 attacks, terrorism – especially in its transnational form – has led many governments to adopt a wide range of surveillance, border control, and military policies, justified by national security concerns [3].

In this broad context, explosives are a key tool of violence:

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substances or mixtures that undergo rapid exothermic reactions, producing heat, gas, and pressure that can be weaponized. Among them, primary explosives are highly sensitive compounds used to initiate secondary charges. They are commonly used in detonators due to their ability to ignite easily with heat, friction, or impact. These materials, though tightly regulated, remain targets for illicit acquisition and misuse in improvised explosive devices (IEDs). Primary explosives are also commonly used directly as the main charge in terrorist attacks, as is the case with triacetone triperoxide (**TATP**), due to their high sensitivity and ease of synthesis from commercially available compounds.

The proliferation of homemade explosives (HMEs) has added another layer of complexity to counter-terrorism efforts. Composed of easily accessible precursors, HMEs/IEDs are attractive due to their low cost and the difficulty of their detection and regulation. The use of IEDs by terrorist groups underscores the strategic shift toward low-cost and readily accessible methodologies used to instill fear and disrupt societies [4].

Terrorism, indeed, has a huge interest in cheap and customizable means to carry out violence. IEDs are typically "low-tech" and assembled from locally available components, frequently incorporating HMEs, including organic peroxides, as the main charge or in initiating secondary explosives [5,6].

Within the taxonomy of explosives, primary explosives are known to be highly sensitive to friction, heat, spark, or impact and are used to initiate secondary charges – a category that includes peroxide-based explosives. Their instability, ease of synthesis from accessible precursors, and lack of industrial or military applications explain their recurrent misuse in HMEs/IEDs [7].

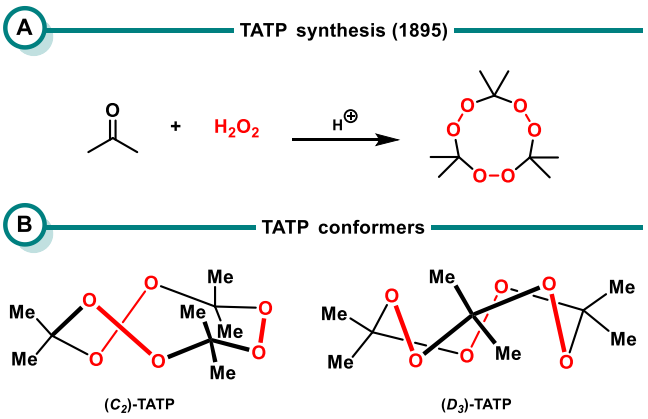
Peroxide-based explosives exemplify these features: they are primary cyclic organic peroxides, typically formed from hydrogen peroxide under catalytic conditions [8], and have been repeatedly associated with terrorist incidents [9–11]. Their detection presents distinct analytical challenges [12–15] because they are prone to sublimation and decomposition and often require indirect assays, via peroxide signatures such as H_2O_2. Consequently, diverse platforms – including ion mobility spectrometry (IMS), (ambient) mass spectrometry (MS), IR/Raman, fluorescence, colorimetric, machine learning, electrochemical methods, and many others – have been developed to address trace detection in vapors, particulates, and environmental matrices. These methodologies, as applied to forensic and explosives science, have been reviewed elsewhere [16–47].

Among these peroxide-based explosives, **TATP** – infamously nicknamed "The Mother of Satan" – is regarded as one of the most dangerous and widely used HMEs. **TATP** is not only extremely sensitive to heat, friction, and electric or mechanical shock, but it has also been repeatedly identified in high-profile terrorist attacks, including those carried out by radicalized individuals using open-source methods and consumer-grade chemicals. Its volatility, instability, and history of misuse make it one of the most feared substances in modern counter-terrorism surveillance.

In the modern history of terrorist use, **TATP** entered broad public awareness after the December 2001 "shoe-bomber" attempt by Richard Reid [48], in which **TATP** served as the initiating charge for a concealed device aboard a transatlantic flight. This widely publicized incident catalyzed intensified aviation security measures and foreshadowed subsequent plots involving peroxide-based explosives. Following 2001, "The Mother of Satan" was used in high-profile attacks in London (2005), Paris (2015), and Brussels (2016), and in subsequent incidents such as in Manchester (2017) [49–51]. Prior to 2001, in 1997, a suicide bombing plot targeting the New York City subway system involving **TATP** was successfully foiled [52].

2. TATP: the Mother of satan

The dreadful nickname "The Mother of Satan" was first coined by terrorist groups and later adopted in official reports [52]. **TATP** was first prepared by Richard Wolffenstein in 1895 by treating acetone with hydrogen peroxide in the presence of an acid catalyst [53]. Wolffenstein reported the formation of a crystalline acetone peroxide that did not form from pure acetone and hydrogen peroxide unless an acid (in his case, a drop of phosphoric acid) was added, indicating an acid-catalyzed condensation process. This discovery highlighted the compound's unique and dangerously unstable chemical configuration, which would later become relevant in the context of IEDs. Despite the limited analytical tools available at the time, he acknowledged the extreme sensitivity and explosive nature of the compound, stating that several detonations occurred during his experiments, albeit without serious injury. He warned that even slight heating could trigger an explosion and emphasized the need for careful handling and suggested storing the compound in sealed capillary tubes to mitigate risks. These early observations, documented in 1895, not only highlight the hazardous properties of **TATP** but also underscored Wolffenstein's cautious yet determined approach to characterizing this highly unstable organic peroxide [53].



Scheme 1. (A) **TATP** was first synthesized in 1895 by Richard Wolffenstein, (B) and now it is known to exist as two main conformers, namely (C_2)-**TATP** and (D_3)-**TATP**.

Subsequent work by Baeyer and Villiger, around 1899-1900, examined the class of acetone peroxides and related superoxides, discussing possible ring sizes and molecular formulas; these studies advanced understanding of acetone-derived peroxides and proposed cyclic oligomeric forms for acetone peroxides [54–56]. The nomenclature and proposals in these classical papers recognized the triacetontriperoxide (tricycloacetone superoxide) among the family of acetone peroxides and discussed polymeric versus cyclic (trimeric) possibilities [54,56]. Definitive structural characterization by modern physical methods was achieved much later: X-ray crystallography studies in the mid-20th century (i.e. 1969) established the cyclic trimeric structure and conformational details that had been hypothesized by early researchers [57].

TATP exists primarily in two conformational isomers (see Scheme 1): a D_3-symmetric structure and a C_2-symmetric structure. Quantum chemical calculations, particularly at the B3LYP/6-311++G(d,p) level of theory, consistently show that the D_3 conformer is thermodynamically favored, being more stable than the C_2 conformer by 0.38 kcal mol^{-1} in the gas phase [58]. This small energy gap suggests a dynamic equilibrium in solution, but with a strong preference for the D_3 form under the conditions used for isolation.

Structurally, the D_3 conformer adopts a cage-like configuration where the three peroxide bridges symmetrically span the three methyl-substituted carbon atoms. This conformation is electrostatically optimized, possessing a zero dipole moment, which further stabilizes it in the gas phase and leads to minimal solvent interactions. The C_2 conformer, in contrast, adopts a less symmetric "open" conformation with one peroxide bridge rotated out of plane, resulting in a higher dipole moment (2.44 D) and increased electronic repulsion.

Calculated structural parameters further support the distinction: in the D_3 conformer, all O–O bonds are equivalent (≈1.475 Å), while the C–O bond lengths (≈1.410 Å) and C–O–O bond angles (≈109.5°) maintain near-perfect tetrahedral geometry. In the C_2 conformer, asymmetry is introduced, with elongation of one O–O bond (≈1.500 Å) and bond angles deviating from ideal tetrahedral values. Natural Bond Orbital (NBO) analysis shows that the D_3 conformer benefits from more uniform nO → σ*CO hyperconjugative interactions, enhancing delocalization.

Experimental support for the existence of both conformers comes from HPLC and NMR studies. HPLC separations in methanol and acetonitrile reveal two distinct peaks for **TATP** conformers [59], with the earlier-eluting peak corresponding to the less stable C_2 isomer. In addition, the conformational equilibrium of **TATP** in solution has been thoroughly examined using a combination of ^{1}H NMR and ^{13}C NMR, all conducted in $CDCl_3$ as the deuterated solvent. These analyses revealed the presence of two major conformers in solution: a dominant D_3-symmetric species and a minor C_2-symmetric species [59].

NMR spectra further differentiate them via distinct chemical shifts: for example, ^{1}H NMR shows singlets at $\delta = 1.47$ ppm (D_3) and 1.42 ppm (C_2), corresponding to the methyl protons of each conformer. These findings indicate that although the D_3 form dominates, the C_2 conformer remains present in detectable proportions in solution, especially under solvent polarity or temperature conditions that modulate the equilibrium. The ^{13}C NMR spectra further support this structural assignment. For the D_3 conformer, a single carbon resonance is observed at δ 25.7 ppm, corresponding to the six equivalent methyl carbons. In contrast, the C_2-symmetric conformer exhibits two distinct carbon resonances due to the loss of perfect equivalence among the methyl groups. The less symmetric geometry produces slightly deshielded chemical shifts in the region of $\delta \approx 26.0$ and 26.5 ppm, attributed to inequivalent methyl carbons adjacent to distorted peroxide bridges. The ^{1}H–^{13}C HSQC spectrum confirmed this correlation by showing one dominant cross-peak for the methyl region. For the minor C_2 conformer, two distinct ^{13}C resonances are observed at approximately δ 26.0 and 26.5 ppm, in agreement with the expected loss of symmetry. ^{1}H–^{13}C HMBC data additionally reveal long-range correlations between the methyl carbons and the quaternary carbon centers within the peroxide ring [59].

Knowing that **TATP** can yield D_3-**TATP** and C_2-**TATP**, whose formation depends mainly on the reaction and recrystallization conditions, an interesting study demonstrated that acid strength, solvent selection, and crystallization rate are the determining factors guiding which conformer predominates [60]. Reactions carried out without strong acid catalysis preferentially afford D_3-**TATP**, the thermodynamically stable conformer. In contrast, the use of strong acids such as HCl, H_2SO_4, or CH_3SO_3H – added to acetone prior to H_2O_2 addition – leads to the rapid formation of C_2-**TATP**, characterized by an opaque appearance. The authors attribute this behavior to a kinetic effect: the strongly exothermic conditions promote fast nucleation and "trap" the higher-energy C_2 conformer before molecular rearrangement to the D_3 form can occur.

Maintaining the reagents at approximately −5 °C and adding H_2O_2 dropwise mitigates violent decomposition and ensures controlled nucleation. Although temperature regulation influences morphology, the decisive parameter remains the acid catalyst employed. Recrystallization plays a central role in isolating pure conformers. When **TATP** is fully dissolved in solvents where it exhibits high solubility – such as in dichloromethane, acetonitrile, or cyclohexane – followed by slow cooling, the resulting clear crystals correspond to D_3-**TATP**. Conversely, recrystallization in methanol, where solubility is limited, favors C_2-**TATP**, producing opaque crystals. The authors report that repeating dissolution at reflux and subsequent cooling gradually converts mixed or opaque samples into the clear D_3 phase [60].

A comprehensive set of analyses employing Raman spectroscopy, X-ray diffraction (XRD), and Differential Scanning Calorimetry (DSC) can differentiate the D_3-and C_2-**TATP** conformers, revealing clear structural and energetic distinctions between the two crystalline forms. Distinct Raman bands confirmed that the clear and opaque crystals correspond to different conformers. Characteristic vibrations appeared at 240, 300-320, 550, and 867 cm^{-1}, indicating variations in peroxide stretching and lattice modes. The isotopic substitution experiments further supported this assignment, as only the band at 867 cm^{-1} remained unshifted upon ^{13}C labeling, demonstrating its association with the peroxide bond. The authors used the displacement of the Raman bands to estimate the interconversion energy between conformers as −4.61 kJ mol^{-1}, consistent with semi-empirical PM3 calculations [60].

The XRD diffractograms revealed distinct sets of Bragg reflections for each crystal type, confirming that the clear and opaque crystals possess different lattice symmetries. These data corroborated the Raman results, showing that D_3-**TATP** (clear) exhibits a more ordered and symmetric crystal structure than C_2-**TATP** (opaque). No diffraction peaks corresponding to residual solvents were detected, confirming the purity of both forms. DSC analysis revealed two distinct melting points, at 88 °C for opaque crystals and 95 °C for clear crystals. The higher melting temperature and narrower range of the clear form confirmed that it corresponds to D_3-**TATP**, the more stable conformer. Mixtures of both forms exhibited two well-resolved endothermic peaks, further validating their coexistence and purity. An in-depth analysis of the synthetic conditions and their variations will be presented in the next section.

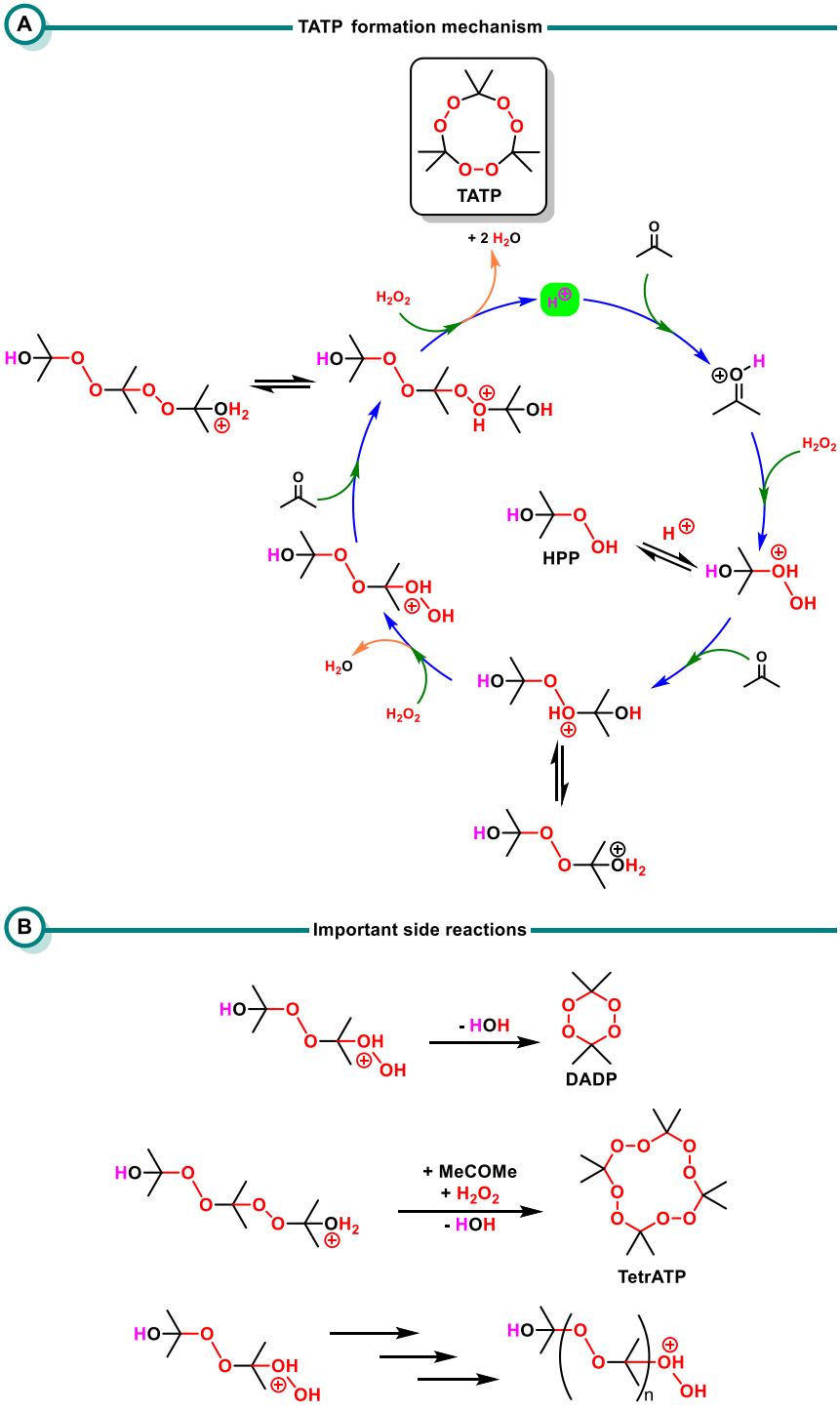
Table 1
Acids used to produce **TATP**.

Acid	References
HCl	[48,61–64]
H_2SO_4	[48,61–63]
HNO_3	[48,61,63]
$HClO_4$	[48,61]
$MeSO_3H$	[48]
BF_3	[48]
CF_3CO_2H	[48]
H_3PO_4	[48]
HCO_2H	[61]
CH_3CO_2H	[61]
Citric acid	[61]
Tartaric acid	[61]
Oxalic acid	[61]
Ascorbic acid	[61]
p-TsOH	[62]
m-Sulfonic acid	[65]
Uncatalyzed	[65]

3. Synthetic parameters, mechanistic pathways, and additional features

Across published protocols, syntheses are predominantly carried out under acidic conditions (Table 1), using a spectrum of Brønsted and Lewis acids to tune rate and selectivity. Representative Brønsted acids include H_2SO_4, HCl, HNO_3, *p*-toluenesulfonic acid (*p*-TsOH), and methanesulfonic acid (MsOH), while common Lewis acids comprise $BF_3{\cdot}OEt_2$, $AlCl_3$, $FeCl_3$, and $TiCl_4$. Practical variables – acid strength and loading, solvent (protic vs. aprotic), temperature control, reagent concentrations, and order of addition – are routinely adjusted to balance conversion, chemoselectivity, and safety. In this review, we compare representative procedures to delineate best-practice windows for reproducible outcomes. Table 1 summarizes acids commonly used to promote **TATP** synthesis. As discussed, yields depend on multiple factors and generally range from as low as 10% to as high as 76%, and are highly sensitive to reaction conditions.

The acid-promoted acetone-hydrogen peroxide system preferentially yields the cyclic trimer (i.e. **TATP**) under the canonical experimental environments reported in the literature. This predominance arises from a combination of mechanistic, kinetic and phase-selection factors.



Scheme 2. Mechanism of **TATP** formation under acidic conditions (A) and selected important side reactions (B).

Mechanistically, electrophilic activation of the carbonyl and sequential formation of peroxy-hemiketal intermediates create a propagation pathway that readily closes to the nine-membered trimer ring. The peroxy-hemiketal sequence therefore provides the most accessible cyclization manifold under protic, acidic media (Scheme 2) [60,65].

Kinetic factors further bias product distribution. Typical syntheses in the literature employ acidic, protic media and reaction/workup regimes in which the sequence of elementary steps that lead to trimer ring-closure proceeds more rapidly than alternative pathways to dimer (**DADP**), tetramer (**TetrATP**) or higher oligomers (Scheme 2). In such regimes, the trimer is the kinetically favored product and is removed from the reacting solution by crystallization, which suppresses further propagation toward heavier oligomers [60,66].

Catalyst identity and medium composition modulate these intrinsic tendencies. Strong Brønsted or Lewis acids accelerate initial activation and can both promote formation of the trimer and, under some conditions, drive secondary transformations (including partial decomposition or conversion between cyclic species). Conversely, alternative catalysts or altered solvation (including differing water content, proticity or specific Lewis acid promoters) can change intermediate lifetimes and thereby shift the product distribution toward **DADP**, tetrameric cycles or other oligomers [60,62,67]. In 2008, an MS study using electrospray

ionization indicated the formation of oligomers at the initial reaction times, which later afford TATP [68].

An important, practical determinant is crystallization and phase selection. The trimer often crystallizes under commonly used isolation conditions, and this solid-phase removal of **TATP** from the reaction mixture prevents onward propagation to larger cyclic species. Differences in recrystallization solvent, catalyst history and thermal treatment are reported to produce differing crystal morphologies and conformers (D_3 versus C_2), and these solid-state outcomes contribute to why the trimer is the most commonly isolated species in forensic and synthetic reports [60,66].

Reports of significant **DADP**, tetramer or even higher oligomer content appear when reaction variables depart from the canonical acidic/protic conditions – for example, with different catalysts, extreme acid regimes, or slow, uncatalyzed pathways – indicating that the product distribution is tunable in principle but that the literature's conventional conditions favor the trimer. These alternative outcomes substantiate that **DADP** and tetrameric products are mechanistically accessible, yet they remain minority products in the bulk of published preparations [65,67,69].

Wolffenstein's original report establishes the earliest experimental observation that concentrated oxidizing media – rather than dilute hydrogen peroxide – are required to produce the acetone "superoxide" [53]. He emphasized that acetone is inert to dilute (≈1%) hydrogen peroxide but is converted to a volatile oxidized product under more concentrated conditions. This observation frames the fundamental requirement for a strongly oxidizing, activating environment for formation of acetone peroxides and indicates that reagent strength and medium composition control whether the ketone is attacked [53].

Later, Baeyer and co-workers (1899) provided corroborating experimental evidence that the oxidation of acetone by strong oxidants yields an immediately formed, crystalline oxidation product whose composition matches Wolffenstein's acetone superoxide [56]. Their characterization confirms that the observed product is a discrete, isolable peroxide-type species rather than an ill-defined mixture, and they attribute its formation specifically to the action of strong, oxidizing reagents such as Caro's reagent (H_2SO_5) [56].

In a follow-up study (1900) [55], Baeyer and collaborators extended these findings by linking the analytical properties of the higher-melting oxidation product to a defined molecularity: molecular-weight determinations and melting behavior indicate a higher-order species. This was interpreted in their work as corresponding to Wolffenstein's trimeric superoxide. The authors thereby advanced the interpretation that under certain oxidizing conditions a specific cyclic oligomer (the trimer) is formed and isolated, and they highlight that the nature of the oxidant (Caro's reagent in their experiments) exerts a decisive, specific effect on product identity [55].

In a subsequent work published in the same year (1900) [54], they adopted a corrective and critical stance: the article questions earlier simplistic assignments of "simple" addition peroxides for many ketone oxidation products and argues that, under the authors' experimental conditions, the formation of such simple peroxides is doubtful. This paper therefore reframes the early literature by warning against over-interpretation of mechanistic assignments based solely on crude analytical data, and it calls for careful molecular characterization to distinguish true cyclic acetone peroxides from other oxidative or persulfuric species [54].

Isopropanol has emerged as a non-obvious but chemically credible precursor for **TATP** formation, with multiple studies demonstrating that 2-propanol can undergo slow oxidative pathways that generate acetone and hydrogen peroxide equivalents under ambient conditions [69–72]. Experimental and forensic evidence shows that aged or improperly stored 2-propanol, particularly when exposed to air, light, and elevated temperature, accumulates peroxides and can yield crystalline **TATP** that is chemically and mechanically identical to laboratory-synthesized material, including high sensitivity to friction and impact. Stoichiometric analyses explicitly note that oxidation of 2-propanol produces acetone and hydrogen peroxide as first-generation products, thereby recreating the classical precursor set required for **TATP** formation without deliberate synthesis. These findings establish isopropanol not merely as a passive solvent, but as an active, time-dependent chemical system capable of generating primary explosives, with clear implications for storage safety, forensic interpretation, and regulatory oversight.

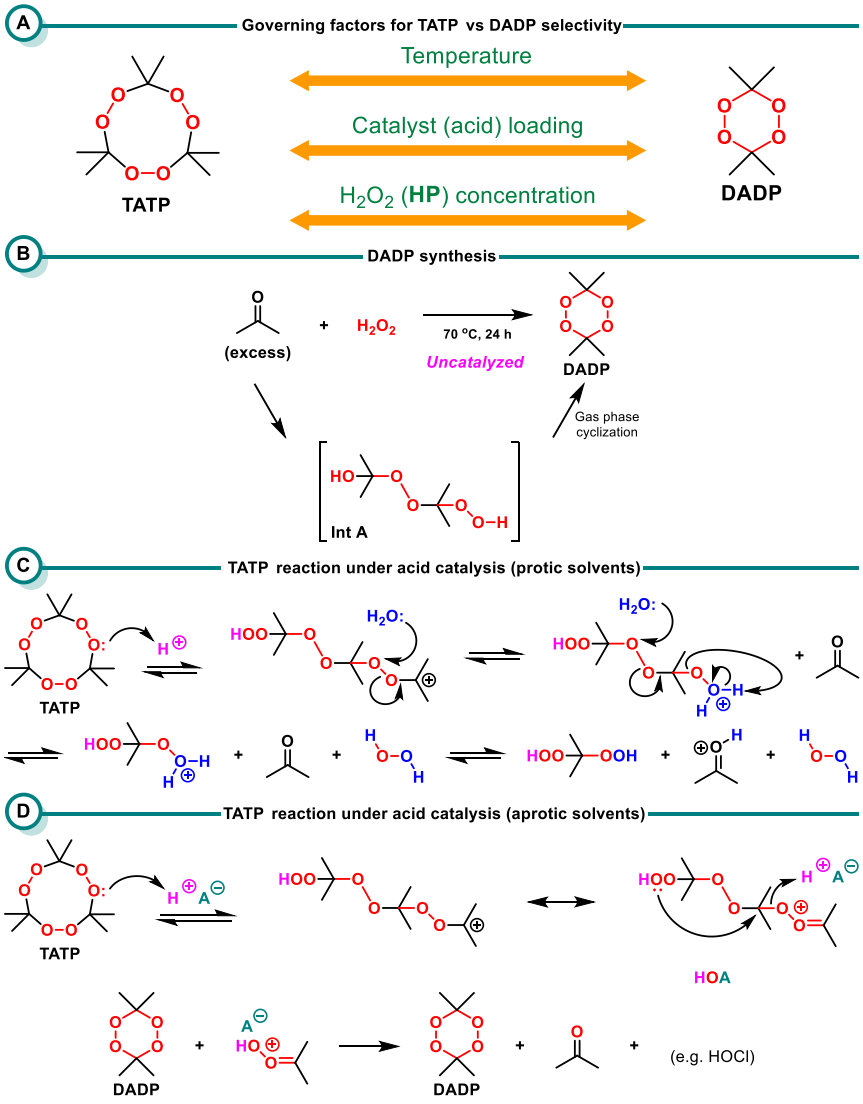
Beyond hydrogen peroxide derived in situ from alcohol oxidation, recent work has demonstrated that alternative oxidizing agents can substitute for hydrogen peroxide in the synthesis of explosive peroxides, broadening the precursor landscape [67]. Sodium perborate and sodium percarbonate were shown to act as effective peroxide donors, enabling the synthesis of **TATP**, **DADP**, and related peroxides with good yields and purity. The justification for considering these materials as alternative peroxide sources is mechanistic: when dissolved in water or treated with acid, they release hydrogen peroxide or peroxide equivalents, yet they are widely available, often unregulated, and commonly used in household and industrial products. This combination of chemical functionality and accessibility creates a significant security gap, as these compounds can bypass controls placed on concentrated hydrogen peroxide while still supporting explosive peroxide formation. These studies underscore that both isopropanol aging and alternative peroxide donors expand the range of chemically plausible initiation pathways for **TATP** formation, reinforcing the need for chemically informed risk assessment rather than precursor lists based solely on traditional reagents.

Still, the classic synthesis of **TATP** under acidic conditions and employing hydrogen peroxide is the dominant methodology used by terrorists. In this context, acid and solvent profoundly reshape the energetics of the initial nucleophilic addition that seeds **TATP** formation. Using M06-2X/6-311+G(2d,p) in SMD water – with HF/6-31G(d) pre-optimizations and solution-phase free energies – a seminal work maps the addition of H_2O_2 to acetone to form 2-hydroperoxy-2-propanol (**HPP** in Scheme 2) and characterizes the relevant twelve conformers and transition states [73]. The calculations support a polar, non-radical pathway; the absence of radical-derived byproducts in prior experiments [74,75] aligns with a nucleophilic mechanism [76], and literature-linked oligomerization then cyclizes to **DADP** or **TATP**, with **HPP** playing a Criegee-like role [77].

Quantitatively, the overall solution free energy for acetone + H_2O_2 → **HPP** is small and slightly endergonic ($\Delta G = +6.3$ kJ mol^{-1} or 1.51 kcal mol^{-1}), while the rate-limiting barrier is highly environment-dependent. In the neutral pathway with no assisting water, the free-energy barrier is ≈ 167 kJ mol^{-1} (39.11 kcal mol^{-1}). Inclusion of catalytic water lowers this to ≈84 kJ mol^{-1} (20.08 kcal mol^{-1}). Under Brønsted acid catalysis, a three-water proton-relay path reduces the barrier further to ≈22 kJ mol^{-1} (5.26 kcal mol^{-1}) – a change consistent with the dramatic accelerations observed experimentally [74].

Mechanistically, water-structured proton transfer is decisive: a network of three water molecules relays a proton from H_2O_2 to the acetone carbonyl, hence stabilizing charge development and compressing barriers along the acid-assisted surface. Together, these results rationalize why strongly acidic, water-rich conditions favor rapid buildup of the early peroxide manifold that ultimately feeds the cyclization steps to **TATP** – not by invoking radicals, but by lowering the polar addition barrier through explicit hydrogen-bonded proton relays.

In seminal work in the early 1970s [74], it was established that acetone and hydrogen peroxide first form the 1.0:1.0 adduct **HPP** in water, with the process governed by general acid-base catalysis and consistent with a polar, non-radical pathway. Equilibrium and kinetic parameters quantified this initial addition and framed the nucleophilic mechanism that precedes oligoperoxide growth. Building on this foundation, subsequent work in the 1970s delineated higher adducts – notably the 1:2 species and condensed adducts – and showed that their formation displays general acid catalysis, with measured rate and



Scheme 3. Factors affecting the synthesis of **TATP**, **DADP** and **TATP/DADP** ratios. (A) Main factors governing the formation of **TATP** or **DADP** (or mixtures). (B) **DADP** synthesis under uncatalyzed conditions, in which the cyclization step occurs in the gas phase. (C) Acid-catalyzed reaction of **TATP** in protic solvents does not favor **DADP** formation. (D) Acid-catalyzed reaction of **TATP** in aprotic solvents (little to no water) favors **DADP** formation.

activation parameters that rationalize stepwise O–O incorporation, and sets the stage for ring closure to cyclic peroxides under appropriate compositions and timescales [75]. Decades later [76], in situ Raman with X-ray confirmation linked these mechanistic insights to operational variables (i.e. pH and temperature) demonstrating how proton activity and thermal regime steer crystallization kinetics and determine which cyclic peroxide precipitates, thereby connecting microscopic adduct chemistry to the macroscopic product.

The Raman kinetic study [76] demonstrated that the reaction between acetone and hydrogen peroxide rapidly forms intermediates, followed by a slower ring-closure step that determines the overall rate. The induction time for **TATP** crystallization decreased markedly with increasing temperature and with acid addition, confirming the catalytic effect of protons on the reaction. Quantitative analysis of induction time versus proton concentration showed that the rate of **TATP** formation is first order with respect to [H^+]. The authors concluded that acid catalysis lowers the energy barrier of the ring-closure step and that increasing temperature up to 313 K (≈40 °C) accelerates the reaction, whereas above 333 K (≈60 °C) **DADP** crystallizes instead of **TATP**, evidencing a thermal threshold for the trimer formation.

It is important to address **TATP/DADP** mixtures, because variations in temperature, acid strength/loading, solvent, and water-to-**HP** (hydrogen peroxide) content not only shift the **TATP/DADP** ratio but also drive post-formation conversion of **TATP** to **DADP**. Depending on conditions, the resulting compositions can range from mixed solids to essentially a single peroxide, as detailed in two articles by the Oxley group (Scheme 3) [78,79].

Oxley and co-workers map the synthesis space and show that temperature, acid strength/loading, **HP** concentration, solvent, and water activity jointly govern whether acetone peroxidation yields predominantly **TATP** or converts it to **DADP** [78,79]. Working with aqueous sulfuric acid, they demonstrate a strong temperature-acidity coupling: under the same stoichiometry (1.0:1.0 mol ratio of 63 wt-% H_2SO_4 to 65 wt-% **HP**), low temperature favors **TATP**, whereas ambient temperature switches the product distribution almost entirely to **DADP**. Specifically, runs at 5 °C gave up to 80% **TATP**, while 25 °C produced ≈100% **DADP**, highlighting a kinetic/thermodynamic pivot controlled by acidity and thermal energy.

They also quantified how **HP** concentration sets the acid threshold required for **TATP** → **DADP** conversion. At a fixed acid strength, more-concentrated **HP** lowers the acid-to-HP ratio needed for complete conversion: with 30 wt-% **HP**, 1.5:1.0 (acid/**HP**) was necessary to obtain

only **DADP**, whereas at 65 wt-% **HP**, 0.8:1.0 sufficed. This observation supports a picture in which increasing oxidant activity and decreasing water content reduce competitive hydrolysis/solvation, enabling the acid-catalyzed ring-opening/rearrangement pathway to **DADP**. Beyond concentration, water content per se is not a simple predictor. The authors caution that water-to-acid ratio alone is insufficient to forecast the product mix because the reaction volume and precipitated solid fraction evolve with composition, complicating phase behavior and mass transfer during crystallization [78].

After the first report [79], the team moves to acetonitrile to dissect mechanism and kinetics. Under strongly acidic, low-water conditions in MeCN, **TATP** is destroyed as fast as it forms, and its acid-promoted decomposition leads to **DADP**, consistent with observed peroxide intermediates (e.g., 2,2-dihydroperoxypropane and 2,2′-dihydroperoxy-2,2′-diisopropylperoxide) that connect formation and conversion pathways.

Water slows both the acid-driven decomposition of **TATP** and its conversion to **DADP**, shifting outcomes toward persistence of **TATP** and incomplete transformation. Similarly, alcoholic media can transiently produce a **TATP**/**DADP** mixture and then stall (e.g., for *t*-butanol). Collectively, these solvent effects show that protic donors and higher water activity damp the cationic/peroxycarbenium pathways (Scheme 3) that funnel **TATP** into **DADP**, while dry, strongly acidic MeCN favors rapid conversion. Overall, these results align with calculations highlighting the roles of water and acid catalysis in the formation of **TATP** [73].

A 2008 short communication established the foundational observation that post-synthesis conversion of **TATP** to **DADP** is catalyst- and temperature-dependent [80]. **TATP** prepared with methanesulfonic, perchloric, or sulfuric acids showed progressive conversion to **DADP** during storage, whereas recrystallized **TATP** or **TATP** prepared with HCl or $SnCl_4$ did not suffer transformation. HNO_3, on the other hand, produced only trace **DADP** after two months. The conversion rate increased with storage temperature and with the molar ratio of catalyst to acetone used during synthesis. Equilibration was observed even at 0 °C, consistent with a residual acidity-driven process in the solid that can be mitigated by recrystallization. Differential thermal analysis and HPLC confirmed the progressive increase of the **DADP** fraction over time, with full conversion observed under certain storage and catalyst conditions.

Following the initial communication on the observed **TATP → DADP** conversion [80], a full paper from the same group quantified these effects and generalized the mechanism in a range of catalysts and temperatures [81]. The kinetics were strongly temperature-dependent – for the same sample, conversion times shifted from 70 to 90 min at 60 °C to 35-65 h at 40 °C and accelerated with higher catalyst loading. The authors also noted catalyst-specific behavior, with H_2SO_4 or methanesulfonic acid giving reproducibly complete conversion, whereas perchloric-acid samples sometimes stalled before full conversion.

An interesting study demonstrated that **DADP** can be synthesized without added acid via a gas-phase pathway (Scheme 3) [65]. The authors argued that the key intermediate **Int A** (see Scheme 3) is dragged into the gas phase by acetone's high vapor pressure and then cyclizes to **DADP** in the presence of excess acetone. They explicitly stated that "*the herein reported formation of **DADP** occurs in the gas phase*" [65], and support this by headspace GC-MS sampling of a sealed system. Quantitative headspace data reinforced this mechanism: acetone and **Int A** dominate the gas phase at ≈36% and ≈60% (by normalized peak area), respectively, consistent with vapor-phase transport of **Int A** preceding cyclization. The authors also note that **DADP** deposits as a white solid on the neck of the flask, consistent with solvent evaporation from the vapor mixture or from exceeding solubility of **DADP** in acetone. Also, they report no **TATP** formation under these uncatalyzed conditions, highlighting a cleaner product profile than acid-catalyzed routes.

Regarding analytical characterization, Raman and FT-IR spectra of the isolated crystals did not show linear open-chain intermediates, consistent with formation of the cyclic product. The GC/MS and DART-MS workflows are described, while cautioning that thermal instability and inlet fragmentation can lower the molecular-ion intensity [65]. The process metrics are explicit: reaction efficiency ≈40% with 80-120 mg isolated in typical runs. High product purity could be inferred from chromatographic peak-area ratios, and the method "is neater" than strong-acid catalyzed syntheses, as judged by the number and area of impurity peaks.

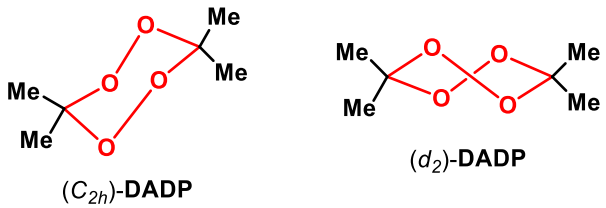


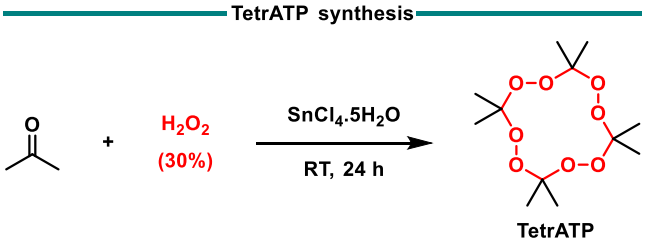
Fig. 1. (C_{2h})-**DADP** (left) and (D_2)-**DADP** (right) conformers of **DADP**.

Mechanistically, the authors link the historical prevalence of **TATP** in aqueous, acid-mediated syntheses to the low aqueous solubility, low polarity, and high vapor pressure of the chain precursors (such as **Int A** in Scheme 3). Uncatalyzed **DADP** emerged when solvent polarity is altered or when **Int A** is transported into the gas phase, as is observed in their setup. Analysis of the headspace composition showed the absence of **TATP,** which is consistent with a gas-phase cyclization route to **DADP** [65].

The conformational landscape of **DADP** has been systematically investigated in the gas phase and in solution to clarify the relative stability and detectability of its chair and twist forms [82]. The study established that gas-phase IR detects **DADP** predominantly in the C_{2h} chair conformation, with <5% of the D_2 twist, and that solution EXSY NMR sets a very low maximum for the twist population, implying $\Delta G > 14.5\ kJ\ mol^{-1}$ (i.e., 3.47 kcal mol^{-1}) at room temperature (Fig. 1). High-level computations that include full thermochemistry and solvation reproduce this stability gap, predicting $\Delta G \approx 14.7\text{-}17.5\ kJ\ mol^{-1}$ (3.51-4.18 kcal mol^{-1}) in polar media. These data collectively argue that the twist conformer lies well above the chair in free energy, under both vacuum and solution conditions.

In terms of mechanistic analysis, the methyl axial-equatorial exchange proceeds via a twist intermediate. EXSY analysis leveraged this pathway to bound the twist population and kinetics. Quantitatively, EXSY limits the twist to ≤0.30% and provides $\Delta G > 14.5\ kJ\ mol^{-1}$ (3.47 kcal mol^{-1}, toluene, 300 K), with a similar lower bound in CD_2Cl_2 at 263 K, reinforcing that twist is rarely populated under typical NMR conditions.

The computational thermochemistry helped explain the energetic gap: the twist conformer has lower vibrational entropy (≈83% of chair), fewer soft modes, and is further destabilized by solvation, together raising ΔG to ≈15-18 kJ mol^{-1} (3.59-4.30 kcal mol^{-1}) depending on method and phase. This framework reconciles the failure to observe twist signals with theory and establishes realistic expectations for detectability. Experimentally, the gas-phase IR spectrum closely matches solid-state data, indicating weak intermolecular interactions in the crystal and validating the chair assignment across phases.



Scheme 4. Synthesis of **TetrATP** catalyzed by $SnCl_4$.

Tetraacetone tetraperoxide (**TetrATP**) typically appears only in trace amounts during the acid-catalyzed peroxidation of acetone that yields **TATP** and/or **DADP**. Multiple studies note the tetramer "in very small amounts," and routine **TATP** preparations often show a minor chromatographic peak assignable to a tetraperoxide species at ≈1-8% of total area [60,65,66].

TetrATP can also be deliberately prepared via the tin-chloride methodology (Scheme 4), which efficiently catalyzes the oxidation of acetone with 30% H_2O_2 at room temperature for 24 h to yield the isolated tetramer confirmed by elemental analysis, FT-IR, NMR, and MS. The yield depends on catalyst loading and composition, reaching up to ≈45% under the reported conditions, whereas added HCl alone gives a low yield [66].

The study [66] also documented practical properties: mp 93-94 °C; unusual stability for an organic peroxide under certain conditions (heating at 120 °C for 4 h or refluxing for 120 h in benzene, chloroform, or light petroleum, including in the presence of V_2O_5, $Co(acac)_3$, or $Fe(acac)_3$); yet it should be handled with great caution because it can be "very explosive" if struck or fired. The authors argue the process is not radical or simple coordination catalysis and appears to proceed through a Lewis acid-catalyzed pathway.

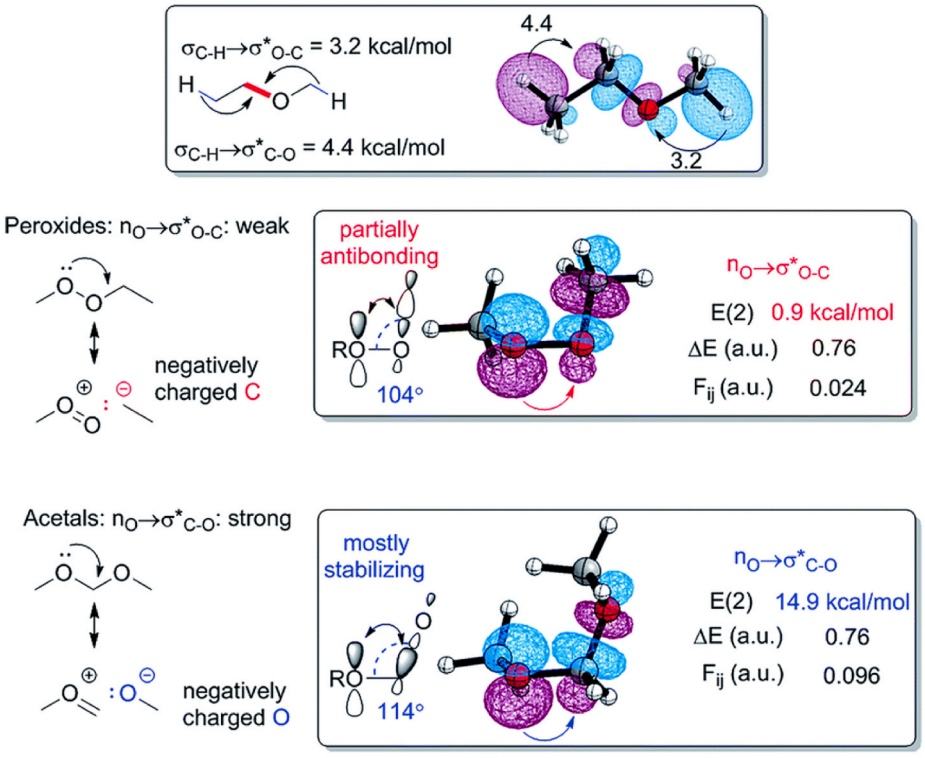
Thermal considerations published later are consistent with the tetramer being intrinsically less stable than the trimer [64]. A structure-reactivity correlation based on O–O bond lengths predicts that the tetramer (if truly obtained) would be roughly two orders of magnitude less stable than **TATP**, reflecting a significantly higher propensity for bond cleavage and thermal decomposition. The same study also reiterates the $SnCl_4$-catalyzed route initially reported and confirms an isolated yield of approximately 45%, lending further support to the proposed synthetic pathway despite the questionable stability of the product.

4. Featured physicochemical properties, analytical principles and detonation of TATP

In the section we will describe the key properties of **TATP,** notably thermal behavior (decomposition kinetics, activation energy, heats of decomposition and combustion, vapor pressure, and thermal signatures from DSC/TGA and laser-heating calorimetry), crystal and polymorphic features and stereoelectronic contributions to instability, environmental behavior and aqueous persistence, detection-relevant metrics (vapor pressure and analytical observables), and performance and safety parameters – reviewing the experimental and thermochemical findings reported across the reviewed literature.

TATP exhibits high-temperature sensitivity, large specific energy release, and gas-phase first-order decomposition kinetics. Comparative manometric studies determined first-order gas-phase decomposition with activation energy 165.8 kJ mol^{-1} (39.6 kcal mol^{-1}) for **TATP** (and **DADP** 159.2 kJ mol^{-1} ≡ 38.0 kcal mol^{-1}) over 140-200 °C, with **TATP** decomposing about twice as fast as **DADP** under identical conditions [64]. The same work measured vapor pressures by a glass Bourdon gauge and provided temperature-pressure data for 75-144 °C, supporting the volatility that underpins vapor-phase reactivity.

Thermochemistry determined by combustion calorimetry and G4 calculations reconciles conflicting historical data: a widely cited 5665 kJ kg^{-1} (1353.97 kcal kg^{-1}) "*heat of explosion*" was called unrealistically high, with reliable measurements for small pressed/cast **TATP** giving ~2803 kJ kg^{-1} (669.93 kcal kg^{-1}), and estimated heats of combustion near 2405 kJ kg^{-1} (574.81 kcal kg^{-1}) [83]. The paper also critiques endothermic enthalpy-of-formation reports and the "*entropic explosion*" hypothesis [84,85], arguing peroxide decomposition is exothermic under realistic conditions [83]. Non-isothermal laser--heating calorimetry (LDTR) provides thermal signatures, initiation and peak reaction temperatures, energy release rate, and total specific energy release for **TATP** at heating rates up to ~60 K s^{-1}, enabling shelf-life assessment and comparison to literature values [86].



Scheme 5. Top: Polarization of σ^*_{C-O} explains why C–O bond is ~40% stronger acceptor at the carbon end. The orbital interactions are quantified by NBO analysis and given in kcal mol^{-1}. Middle and bottom: The difference between $n_O \rightarrow \sigma^*_{C-O}$ interactions in acetals and $n_O \rightarrow \sigma^*_{O-C}$ interactions in peroxides is amplified >16-fold (>1600% difference!). Scheme 5 has been reproduced from Ref. [94]. The article is open access and licensed under a Creative Commons Attribution.

Reviews on detection in water and updated vapor-pressure compilations emphasize the very low volatility of these cyclic peroxides (i.e. **TATP** and/or **DADP**) and the analytical implications for environmental and security monitoring [7,87]. It is important to remember that, although its vapor pressure is low, it can be sufficient to reach saturation in an enclosed space and, therefore, to cause an explosion simply by switching on a light. This is why many describe the "spontaneous" explosion of **TATP**.

Interestingly, LDTR established both initiation and peak reaction temperatures for the material and quantified its specific energy-release rates, providing a clear thermal profile of the decomposition process [86]. The study further applied rigorous kinetic/model-selection procedures appropriate for fast, strongly exothermic reactions to identify the most representative reaction model and parameters. Together, these results define the onset of hazard and the maximum thermal output expected under runaway conditions. Importantly, LDTR also reports no measurable degradation of **TATP** over a three-year period under the non-ideal storage conditions tested – no significant loss of energetic content, no detectable change in decomposition onset, and no observable alteration to the thermal signature – suggesting the compound's shelf life remained effectively unchanged in those particular conditions [86].

Thermogravimetric/DSC work on **TATP** (including thermolysis in solvents and non-isothermal analysis) addressed onset temperatures, kinetics parameters, and coupled phase-transition/decomposition behavior [88–93]. TG, DSC, and solvent-assisted thermolysis studies provide a coherent picture of onset temperatures, kinetic parameters, and coupled phase-transition/decomposition behavior of **TATP** [88,90,91,93].

TATP shows single-stage rapid mass loss under TG conditions due to its volatility, with decomposition initiating between 40 and 100 °C depending on purity and synthesis conditions [88,90,91,93]. DSC consistently identifies low exothermic onset temperatures (~40–60 °C in acid-catalyzed samples; ~95 °C for some recrystallized or mildly catalyzed preparations) and large decomposition enthalpies, often exceeding 2500-2700 J g^{-1} [90,91]. These characteristics reflect the energetic nature of the O–O bonds and the rapid gas evolution accompanying decomposition.

Kinetic parameters obtained by non-isothermal DSC methods reveal activation energies ranging from ~65 kJ mol^{-1} (15.53 kcal mol^{-1}) (acid-synthesized samples) to ~155 kJ mol^{-1} (37.04 kcal mol^{-1}) (methanol-recrystallized samples), demonstrating the strong dependence of kinetic behavior on sample history and purity [90]. These multi-rate DSC studies also demonstrate coupled phase-transition/decomposition behavior, where melting or sublimation overlaps with decomposition exotherms, complicating kinetic interpretation [93].

Thermolysis of cyclic peroxides structurally related to **TATP** in solution follows pseudo-first-order kinetics, with solvent-dependent rate enhancements linked to preferential solvation and stabilization of diradical intermediates [89]. Although the cited study focuses on acetone-derived di- and triperoxides rather than neat **TATP**, it directly supports the mechanistic expectation that solvent polarity and hydrogen-bonding interactions modulate activation barriers and decomposition rates during peroxide thermolysis. These TG/DSC and solution-phase thermolysis results define **TATP** as a low-onset, highly exothermic, kinetically labile peroxide whose decomposition is strongly modulated by thermal program, synthesis route, phase behavior, and solvent environment.

The stereoelectronic origins of **TATP** instability arise from the inherent weakness of the O–O σ-bonds and the highly strained cyclic peroxidic framework. Quantum-chemical studies demonstrate that initiation pathways are dominated by σ* orbital population and facile homolysis, with low activation energies associated with peroxide bond cleavage (Scheme 5) [94]. These features align with the "*entropic explosion*" model [84], in which rapid gas release and highly exergonic fragmentation channels drive detonation once the first O–O bond ruptures. Theoretical analyses also show that electron donation into σ* orbitals and repulsive lone-pair interactions destabilize the trimeric cage, providing a stereoelectronic basis for its extreme sensitivity.

Polymorphism strongly influences the solid-state behavior of **TATP**. At least six distinct polymorphic forms have been identified, and their formation depends on both the acid catalyst and the crystallization solvent. Although all polymorphs share the same molecular conformation with D_3 point symmetry, they differ markedly in packing, disorder, and unit-cell parameters. Such variations in crystal packing are particularly important because they directly affect the material's explosive properties [95]. This knowledge is essential, as these solid-state variations provide a mechanistic foundation for understanding how sensitivity and thermal behavior differ among the polymorphs and help explain differences in calorimetry measurements.

Additional insight into how intermolecular interactions shape peroxide-containing crystals comes from studies of water-cluster inclusion in structurally related supramolecular lattices. The hydrated solid forms a sodalite-like **TATP** porous supramolecular architecture that captures a bigger-sized heptadecameric water cluster guest [96]. Although this study does not involve **TATP**, it demonstrates how hydrogen-bonded water networks interact within the crystalline phase and can alter packing, symmetry, and stability. These results provide a conceptual basis for understanding how trace water or solvent inclusions could modulate peroxide crystal behavior.

Within the broader classification of energetic materials, the structural attributes of peroxides differ fundamentally from classical nitro-based explosives [97]. This classification of energetic crystals establishes that energetic-crystal analysis focuses on correlations between molecular structure, packing, and reactivity. Combined with the properties described above – weak O–O σ bonds, high entropic output upon decomposition, and packing-dependent polymorphism – **TATP** aligns with low-density, highly strained molecular explosives whose performance is dominated by peroxide bond chemistry rather than nitro-group energetics [97].

Vapor-pressure data provided a quantitative basis for assessing the feasibility and limitations of vapor-phase detection of peroxide explosives. A recent comprehensive review emphasizes that "*the vapor pressures of many explosive compounds and illicit drugs are extremely low relative to volatile organic compounds*" and that determining accurate values at ambient temperature is challenging, often requiring extrapolation from higher-temperature measurements [87]. This work compiles experimentally measured vapor pressures at 25 °C and highlights that, for peroxides such as **TATP** and **DADP**, the available vapor is intrinsically limited, constraining detection strategies based on equilibrium vapor concentrations. For **TATP** specifically, the updated review reports that multiple independent measurements at or near ambient temperature converge to very low vapor pressures, despite some variability arising from synthesis conditions and polymorphism. The authors note that "*there are seven new reported vapor pressures for **TATP***" and, after critical evaluation, derive an average value at 25 °C on the order of 10^{-5} atm [87]. The review further emphasizes that, owing to their very low vapor pressures, trace identification generally depends on the detection of solid residues deposited on surfaces rather than on free vapor, since dilution effects and surface interactions can decrease vapor-phase concentrations by several orders of magnitude relative to equilibrium values.

Complementary high-temperature studies provided the experimental basis for extrapolation to ambient conditions. A comparative thermochemical investigation determined the vapor pressures of **TATP** and **DADP** using a glass Bourdon gauge over elevated temperature ranges, reporting that the vapor pressures of **TATP** and **DADP** were determined over the temperature range 75-144 °C and 67-120 °C, respectively [64]. These measurements enabled determination of sublimation and evaporation parameters and revealed inflection points at the melting transition, which are essential for reliable Clausius-Clapeyron extrapolation.

Environmental stability studies also demonstrated that peroxide-based explosives such as **TATP** and **DADP** can persist as detectable residues in aqueous and alcoholic matrices, but their stability is strongly dependent on storage conditions, temperature, and light exposure [98]. Systematic experiments examining residues on alcohol wipes, in 60% (*v/v*) methanol/water extracts, and on glass surfaces showed that post-collection degradation is a critical factor for forensic and environmental interpretation. The authors highlight that the persistence of trace explosives on collected debris and on sampling substrates after collection represents a critical element of the overall analytical workflow. In the specific case of **TATP**, they demonstrated that storage under dark conditions and at low temperatures is necessary to limit degradation and material loss over time [98]. The overall findings indicated that aqueous and alcoholic environments can preserve **TATP** residues sufficiently for delayed analysis, provided that volatilization and photodecomposition are minimized.

From an environmental and surveillance perspective, wastewater has emerged as a key matrix for tracing clandestine synthesis of peroxide-based explosives. A focused review points out that, in the preparation of peroxide-based explosives, substantial amounts of precursor compounds or explosive residues can be released into wastewater and subsequently reach natural water systems [7]. Although **TATP** exhibits low vapor pressure and limited water solubility, its partial dissolution and transport in aqueous systems make it a viable chemical marker when monitored close to the source. The review additionally remarks that wastewater has been shown to be an effective indicator matrix for early analyte detection [7], and it extends this approach to peroxide-based explosives and their precursor compounds.

Additives and mitigation strategies have been explored as practical routes to reduce the sensitivity and thermal hazard of **TATP**, with ionic liquids and surfactants emerging as particularly effective modifiers of its decomposition behavior. Experimental calorimetric studies show that incorporation of ionic liquids such as 1-butyl-3-methylimidazolium tetrafluoroborate ($[C_1C_4.Im][BF_4]$) lowers the maximum decomposition temperature and reduces the overall heat release of **TATP**. A comprehensive DSC/TG/VSP2 study showed that increasing the concentration of $[C_1C_4.Im][BF_4]$ led to reductions in both the maximum peak temperature (T_{max}) and the mean decomposition enthalpy of **TATP** [99]. These results indicate that ionic liquids can attenuate thermal runaway and pressure rise, thereby mitigating explosion hazards during handling and processing.

Beyond thermal moderation, ionic liquids and surfactants also provide pronounced mechanical desensitization. A focused investigation addressing safe sampling and formulation demonstrated that dissolving peroxide explosives in ionic liquids markedly lowers their sensitivity to friction and impact. The study reports that even highly sensitive peroxide-based explosives can be safely handled when dissolved in ionic liquids, and that these systems display long-term stability while remaining suitable for forensic analysis [100]. The same work attributes this stabilization to surfactant-like interactions between the ionic-liquid alkyl chains and **TATP** crystals, combined with electrostatic effects that protect crystal edges from mechanical stress, providing a mechanistic explanation for the observed desensitization [100].

From a process-safety and infrastructure perspective, the hazards associated with **TATP** extend beyond laboratory-scale handling to potential large-scale impacts on industrial facilities. Quantitative risk and vulnerability analyses demonstrate that improvised explosives containing **TATP** can trigger severe domino scenarios, even at relatively modest charge sizes. It is explicitly stated that "***TATP is a primary explosive … very sensitive to impact or friction***" and that attacks using **TATP** can generate blast waves capable of damaging process equipment and initiating escalation sequences [101]. These findings underscore the necessity of integrating security management with conventional process-safety frameworks when assessing risks related to peroxide explosives.

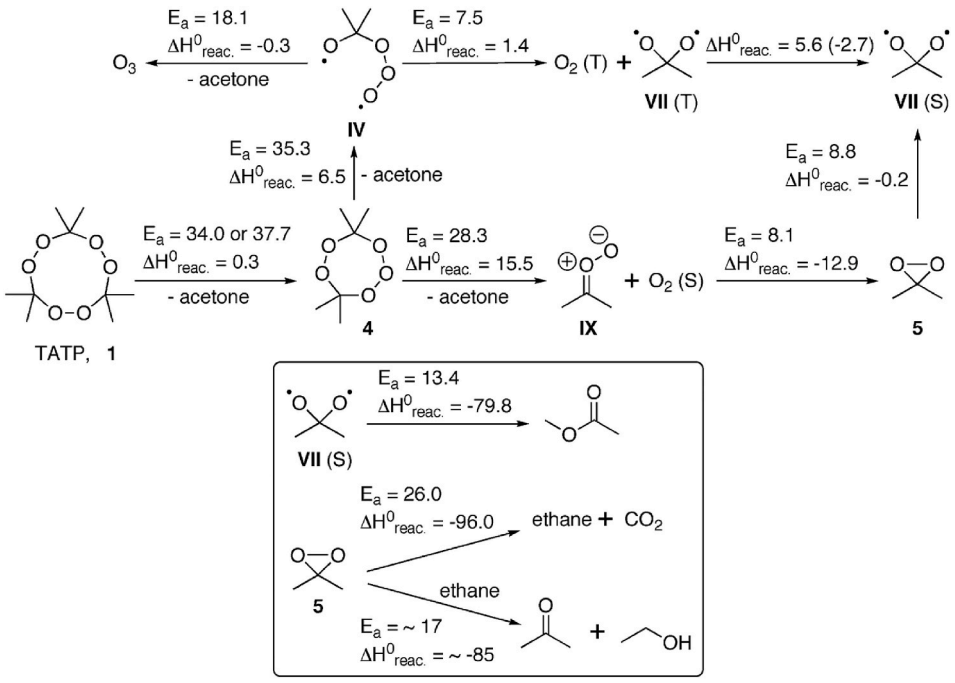
Emergency-response strategies have also been evaluated using calorimetric approaches to identify suitable fire-extinguishing and hazard-suppression measures. DSC and TG studies examining mixtures of **TATP** with common extinguishing agents reveal that certain additives can significantly delay decomposition and reduce heat release. In particular, it was shown that "*adding KBC, ABC, XBC, and BC powders can reduce the heat of decomposition and delay the decomposition temperature of **TATP***" and that pure **TATP** decomposed rapidly and final decomposition was calculated before 200 °C with a heat of decomposition of approximately 2500 J g^{-1} [102]. Such calorimetric data provide quantitative parameters relevant for emergency response planning, highlighting which suppression agents may mitigate, rather than exacerbate, thermal hazards during fire scenarios.

Performance and detonation metrics of **TATP** have been quantified through systematic studies of mixtures with oxidizers and comparative blast analyses against TNT. Early work on **TATP**/AN (AN = ammonium nitrate), **TATP**/AN/W (W = water), and **TATP**/UN (UN = urea nitrate) mixtures established that relative performance (RP), measured by ballistic mortar, depends strongly on composition, oxygen balance, and water content. It was demonstrated that the RP of the mixtures and of their individual constituents, benchmarked against TNT, was evaluated using ballistic mortar measurements, and that, as a function of water content, **TATP**/AN formulations can exhibit detonability values exceeding those of ANFO (ammonium nitrate/fuel oil) explosives [103]. These results demonstrate that blending **TATP** with ammonium nitrate can substantially enhance or suppress performance relative to neat **TATP**, depending on formulation. A complementary study directly correlated thermal reactivity with detonation behavior in peroxide-containing mixtures. For **TATP**/AN and **TATP**/UN systems, a correlation with the squared values of the calculated detonation velocities, indicating that this trend mirrors the fundamental chemical micro-mechanism governing detonation in these mixtures was observed [104]. These findings link low-temperature thermolysis pathways to macroscopic detonation performance, reinforcing that initiation efficiency in **TATP**-based mixtures is governed by the same peroxide bond cleavage processes that dominate thermal decomposition.

Using ballistic mortar tests, the power of various formulations was assessed, showing that the ternary mixtures of **TATP**/AN/W have relatively high power in case of the low water contents, while power decreases significantly with increasing the water content in the mixture to more than 30% [105]. These data confirm that **TATP** can deliver substantial mechanical work, but that its performance is highly sensitive to dilution and formulation.

Direct blast measurements have provided a quantitative comparison between **TATP** and TNT under controlled conditions. Experimental investigations using small explosive charges showed that key blast wave characteristics – namely incident overpressure, impulse, and positive-phase duration – were measured for 50 g samples of dry **TATP**. From these data, TNT equivalency values of approximately 70% based on overpressure and 55% based on the positive-phase impulse were determined [106]. Independent assessments of improvised energetic materials likewise stress that TNT equivalence is metric-dependent, indicating that equivalence values derived from peak blast overpressure are higher than those obtained from blast impulse measurements [107]. Overall, these studies place **TATP** below TNT in overall brisance and impulse, while still confirming its significant blast capability and efficient initiation despite its low density.

A complementary perspective is provided by a theoretical study that examines **TATP** as an energetic scaffold whose intrinsic limitations could be overcome through rational molecular design [108]. Using DFT calculations at the B3LYP/6-311++G(d,p) level, the authors evaluated a series of **TATP** derivatives bearing electron-withdrawing and electron-donating substituents (–CF_3, –NO_2, and –NH_2), correlating O–O bond lengths, electronic structure, crystal density, and heats of formation with predicted detonation performance. The results indicate that the weak O–O bonds characteristic of **TATP** underpin both its high sensitivity and poor energetic output, but that appropriate substitution



Scheme 6. Possible intermediates and final products of the thermal decomposition of **TATP**. The scheme includes the activation energy and reaction enthalpy for each step relative to **TATP**. Reproduced with permission from Ref. [84]. Copyright 2005, American Chemical Society.

can shorten O–O bond lengths, increase density, and significantly enhance calculated detonation velocity and pressure. In particular, nitro substitution yields a derivative predicted to outperform both **TATP** and TNT in detonation parameters, highlighting that the peroxide ring itself can serve as an energetic skeleton when embedded in a more favorable electronic and structural context [108].

5. Detonation mechanisms, deactivation of TATP and safety

Fundamental detonation initiation mechanisms of **TATP** have been examined through combined experimental, theoretical, and thermal analyses, converging on a picture in which initiation is governed by the intrinsic lability of the peroxide framework rather than by classical hot-spot mechanisms typical of nitro-based explosives. Molecular-level studies demonstrate that initiation begins with homolytic cleavage of an O–O bond, producing highly reactive radical intermediates that rapidly propagate decomposition through concerted fragmentation pathways [109]. This bond rupture is facilitated by the weak σ O–O bond and by stereoelectronic factors that destabilize the cyclic peroxide cage, lowering the energetic threshold for initiation under thermal, mechanical, or electrostatic stimuli.

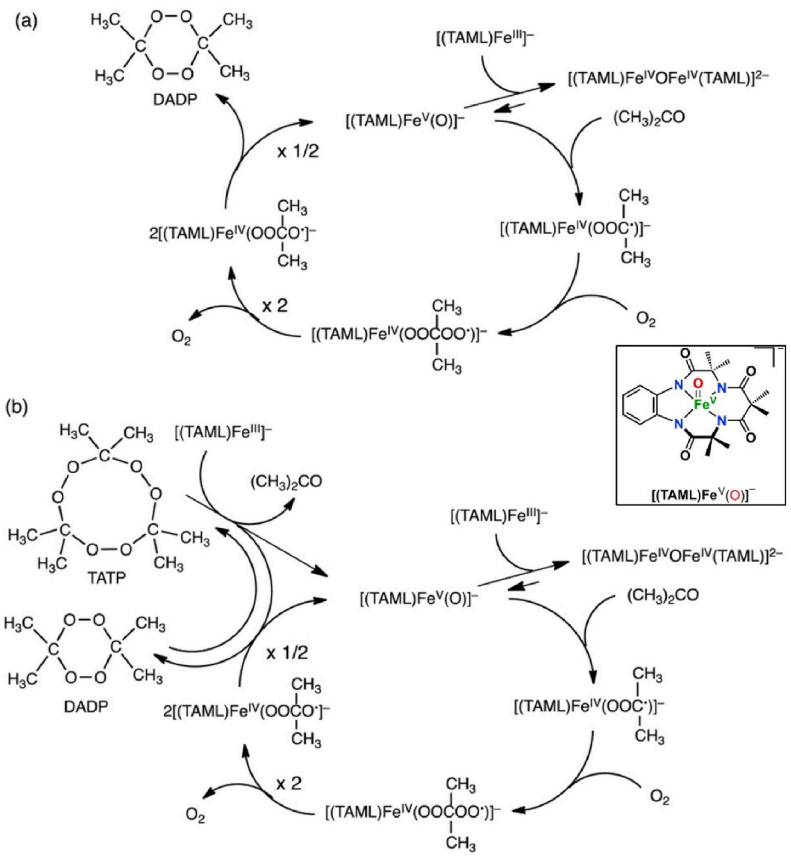
A defining feature of **TATP** detonation is the dominant role of entropy rather than enthalpy in driving the explosive event. The decomposition of a condensed-phase **TATP** molecule into multiple small, gaseous products produces a large positive entropy change, which strongly favors rapid expansion once bond cleavage is initiated. This behavior underpins the "entropic explosion" concept, in which detonation is driven by the sudden generation of a high number of gas-phase species rather than by exceptionally high heats of formation [84]. As a consequence, even modest local excitation can trigger runaway decomposition if confinement is sufficient to couple entropy generation with pressure buildup. Scheme 6 shows a plausible mechanism, based on theoretical calculations and experimental evidence, to describe the thermal decomposition of **TATP** [84].

Thermal analysis provides complementary insight into how initiation thresholds are reached under controlled heating. Detailed kinetic studies reveal that **TATP** undergoes a single dominant decomposition event, with activation energies that are low relative to conventional secondary explosives and that vary with sample history and preparation conditions [110]. These results indicate that initiation does not require extensive preheating or phase transitions but can proceed directly once critical bond cleavage occurs, consistent with the absence of stabilizing intermolecular interactions in the crystal lattice.

Importantly, mechanistic studies of deliberate **TATP** destruction clarify the narrow boundary between controlled decomposition and accidental initiation. Chemical or thermal interventions that promote selective O–O bond cleavage without rapid gas accumulation can neutralize **TATP**, whereas conditions that allow radical propagation and gas evolution to couple in confined environments favor detonation [48]. Together, these studies establish that **TATP** detonation is fundamentally governed by peroxide bond instability, entropy-driven gas formation, and minimal kinetic barriers, distinguishing its initiation mechanisms from those of more structurally robust energetic materials.

Phase behavior and polymorphism have been shown to influence the physical properties of **TATP** and its sensitivity. As discussed earlier, crystallographic studies identify two crystalline modifications, commonly denoted D_3 and C_2, which differ in molecular symmetry and crystal packing, demonstrating that **TATP** does not exist as a single, structurally invariant solid [60]. These structural differences establish that the solid-state form of **TATP** depends on crystallization history rather than solely on molecular composition. Process and crystallization studies further demonstrate that distinct polymorphs of **TATP** can be obtained by varying experimental conditions such as solvent and temperature, confirming the polymorphic nature of this peroxide explosive [111]. These observations show that **TATP** can adopt different solid forms under accessible laboratory (or improvised) conditions, which has implications for reproducibility and handling.

Solid-state transformations between peroxide species have also been documented. Independent studies report that **TATP** can convert to **DADP** under specific conditions, including heating and storage, indicating that chemical composition and phase behavior may evolve without complete decomposition [80,81]. This transformation represents a compositional change that occurs within the solid state and alters the identity of the peroxide present. All these solid-state properties are factors capable of affecting the propensity of **TATP** to detonate, as multiple variables may be present within a given sample of this solid.



Scheme 7. (a) Autocatalytic mechanism for the formation of $[(TAML)Fe^{V}(O)]^{-}$ and **DADP** in acetone. (b) Autocatalytic mechanism for the formation of $[(TAML)Fe^{V}(O)]^{-}$ via **TATP** in acetone. Reproduced with permission from Ref. [115]. Copyright 2017, Royal Society of Chemistry.

Considering the reported cases of "spontaneous detonation" associated with **TATP**, these aspects should be taken seriously into account.

Volatility and sublimation are additional factors governing the physical behavior of **TATP** and may directly influence its detonation. Experimental measurements show that **TATP** sublimes at ambient temperature, leading to measurable mass loss and changes in sample properties over time [61]. These findings establish that **TATP** exists in dynamic equilibrium between solid and vapor phases, which can influence the condition of solid samples during storage.

Gas-phase processes represent a critical intermediate between the solid-state behavior of **TATP** and macroscopic detonation phenomena. Quantitative studies of vapor-phase behavior demonstrate that both **TATP** and **DADP** establish measurable gas-phase concentrations under ambient conditions, confirming that sublimation leads to a continuous exchange between condensed and gaseous phases [112]. These measurements show that vapor pressures and resulting concentrations depend on temperature and compound identity, establishing that the presence of **TATP** vapors is an intrinsic physicochemical property rather than an artifact of handling.

The existence of a gas-phase population has implications for predetonation behavior, particularly in confined or semi-confined environments. Gas-phase accumulation can contribute to heterogeneous initiation scenarios in which vapor-phase molecules participate in early-stage decomposition or ignition processes prior to bulk solid involvement. The comparative behavior of **TATP** and **DADP** in the gas phase further indicates that solid-state transformations altering composition can modify vapor-phase exposure and associated risks [112].

Blast experiments provide a macroscopic manifestation of how these physicochemical properties translate into detonation outcomes. Controlled detonation studies show that **TATP** generates blast waves with measurable incident overpressure, impulse, and positive-phase duration, enabling direct comparison with TNT under identical charge conditions. These results demonstrate that, despite its low density and non-nitro composition, **TATP** produces substantial blast effects once detonation occurs, confirming that pre-detonation processes such as sublimation and vapor-phase behavior do not diminish its ability to generate high-pressure shock fronts.

Chemical deactivation and controlled decomposition of **TATP** have been investigated to delineate conditions under which peroxide bond cleavage proceeds without transition to detonation. Solution-phase studies demonstrate that **TATP** exhibits limited stability in common solvents and that its persistence depends on solvent identity and conditions, indicating that chemical environment strongly modulates decomposition pathways [113]. These observations establish that controlled degradation can occur in liquid media without immediate energetic release, provided that gas evolution and radical propagation remain constrained.

Redox-active species have been shown to play a decisive role in promoting non-detonative decomposition. Studies examining iron salts and thiols report that Fe(II)/Fe(III) couples facilitate peroxide bond cleavage through electron-transfer processes, leading to fragmentation products without the rapid pressure buildup characteristic of detonation [114]. Complementary work demonstrates that iron-mediated oxidation not only accelerates **TATP** breakdown but can also promote interconversion between peroxide species, including the formation of **DADP** under oxidative conditions Scheme 7 [115].

Broader surveys of metal-ion reactivity confirm that **TATP** is susceptible to degradation in the presence of specific cations, with reaction outcomes depending on metal identity and coordination chemistry rather than on nonspecific heating effects [116]. These findings indicate that chemical deactivation can be rationally tuned by selecting reagents that selectively weaken O–O bonds while avoiding conditions that favor

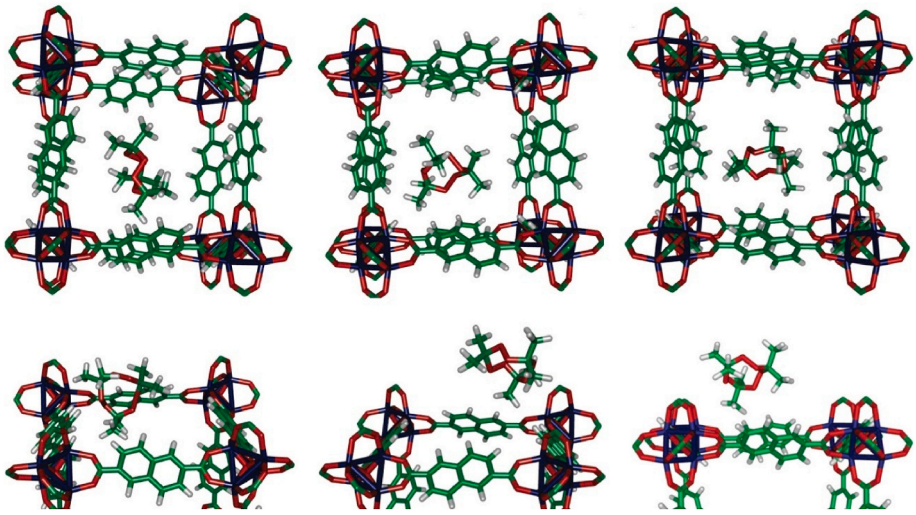


Fig. 2. (Top) **TATP** stable configurations in the interior of IRMOF-8 cage at T = 300 K (26.85 °C). (Bottom) **TATP** stable configurations on the surface of IRMOF-8 at T = 300 K. Reproduced with permission from Ref. [118]. Copyright 2010, American Chemical Society.

runaway gas-phase chemistry.

Highly aggressive oxidative systems highlight the boundary between neutralization and hazard. Investigations involving piranha-type mixtures demonstrate rapid and complete destruction of **TATP**, emphasizing that extreme oxidizing environments eliminate the peroxide framework but pose substantial secondary risks associated with heat release and reagent reactivity [117].

Materials-assisted mitigation strategies address **TATP** hazards by exploiting specific interactions between the peroxide molecule and solid surfaces, enabling adsorption, partial deactivation, or controlled detection without bulk energetic release. Early studies on porous crystalline materials demonstrate that metal-organic frameworks (MOFs) can interact with **TATP** through host-guest mechanisms (Fig. 2), in which pore size, surface chemistry, and framework topology govern adsorption behavior [118]. These results establish that confinement within ordered porous matrices can immobilize **TATP** molecules, altering their local environment and potentially reducing mobility and sensitivity.

Polymeric materials designed with molecular recognition motifs provide a complementary mitigation approach. Molecularly imprinted polymers (MIPs) tailored for **TATP** exhibit selective binding based on size, shape, and functional-group complementarity, demonstrating that synthetic polymer networks can discriminate **TATP** from structurally related compounds [119]. Such specificity indicates that adsorption is not purely nonspecific physisorption but arises from designed binding sites that stabilize the peroxide within the polymer matrix, supporting applications in selective capture and passive mitigation.

Metal oxide surfaces further illustrate how material-assisted interactions influence **TATP** behavior. Studies employing TiO_2-based systems show that **TATP** can be detected and decomposed at oxide surfaces under controlled conditions, highlighting the dual role of inorganic materials as both sensing platforms and active interfaces that promote peroxide bond disruption [120]. These findings confirm that surface-mediated processes can engage the peroxide framework directly, offering routes to mitigate or monitor TATP through heterogeneous interactions.

Analytical signatures associated with **TATP** detonation and deactivation provide molecular-level evidence of peroxide integrity, transformation, and breakdown pathways. Mass spectrometric investigations demonstrate that **TATP** and **DADP** generate characteristic ionic species that enable unambiguous differentiation between intact peroxide explosives and their degradation products [121].

Integrated safety and countermeasure considerations synthesize mechanistic understanding, materials behavior, and operational constraints to address the risks posed by **TATP** across handling, storage, and response scenarios. Experimental studies focusing on mechanical sensitivity demonstrate that friction sensitivity of **TATP** can be reduced under specific conditions, confirming that sensitivity is not immutable but depends on physical form and external modifiers [122]. These findings establish that targeted interventions can lower the probability of accidental initiation, while also highlighting that such reductions do not eliminate the intrinsic energetic character of the peroxide.

At the facility and systems level, process-safety analyses emphasize that **TATP** hazards cannot be managed solely through chemical knowledge, but require integrated risk assessment frameworks. Modeling studies addressing industrial and laboratory environments show that explosive peroxide scenarios demand consideration of layout, confinement, ventilation, and emergency response planning, as even small quantities can generate severe consequences if initiation occurs [101]. These analyses position **TATP** as a material for which conventional safety assumptions based on nitro-based explosives are insufficient.

Together, these works demonstrate that effective countermeasures against **TATP** rely on combining molecular-level insights with engineering controls. Reducing mechanical sensitivity, limiting confinement, controlling environmental conditions, and implementing informed emergency strategies are complementary elements of a unified safety approach. This integrated perspective underscores that mitigation of **TATP** risks depends on coordination between chemical deactivation strategies, materials handling protocols, and system-level safety design.

6. Analytical methods to identify and quantify TATP

Analytical identification of **TATP** remains challenging due to its chemical structure, physicochemical properties, and behavior under environmental conditions. Unlike nitro-based explosives, **TATP** lacks strong chromophores or ionizable functional groups, which limits the applicability of conventional spectroscopic and electrochemical detection methods and necessitates indirect or highly specialized analytical approaches [123]. Its high volatility and tendency to sublime at ambient temperature further complicate sampling, transport, and quantification, particularly in trace analysis and post-event scenarios [87].

Mass spectrometry-based techniques provide high selectivity and sensitivity for **TATP** and its degradation product **DADP**, enabling differentiation between intact peroxide explosives and transformation products. However, these methods often require laboratory-based instrumentation and controlled ionization conditions, which limit

Table 2
Summary of analytical methods reported for the detection and quantification of **TATP**.

Entry	Detection method	Observation	Detection	Year	Ref.
1	GC-MS, FTIR	-	Direct	1986	[125]
2	GC-MS, FTIR	-	Direct	1992	[126]
3	Colorimetry	A hand-held photometer was used for on-field detection; non-selectively sensitive to HMTD	Indirect (H_2O_2)	2002	[127]
4	HPLC-APCI-MS	-	Direct	2002	[128]
5	HPLC-FL	TATP and HMTD were decomposed post-column prior to detection	Indirect (H_2O_2)	2003	[129]
6	Spectrophotometry	UV decomposition (254 nm) followed by reaction with 2,2′-azino-bis(3-ethylbenzothiazoline)-6-sulfonate	Indirect (H_2O_2)	2003	[130]
7	NMR, Raman and IR spectroscopy, IMS, IMS-MS	First report on the use of IMS for TATP detection	Direct	2003	[131]
8	HS-GC-MS	Analysis of post-explosion debris	Direct	2004	[132]
9	GC-MS	Analysis of post-explosion debris by solid phase microextraction	Direct	2004	[133]
10	SIFT-MS	Products of the gas-phase reaction between TATP and H_3O^+, O_2^+ and NO^+ were analyzed	Direct	2006	[134]
11	SPI-MS	-	Direct	2006	[135]
12	DESI-MS	-	Direct	2006	[136]
13	LI-MS	-	Direct	2006	[137]
14	GC-MS	Best results using ammonia positive ion chemical ionization	Direct	2006	[138]
15	IMS	-	Direct	2008	[139]
16	Fluorescent sensor	-	Indirect (H_2O_2)	2008	[124]
17	Fluorescent sensor	-	Indirect (H_2O_2)	2008	[140]
18	GC-MS	Analysis of post-explosion debris by solid phase microextraction	Direct	2008	[141]
19	ESI-MS	Also identified synthesis intermediates	Direct	2008	[68]
20	Terahertz spectroscopy	-	Direct	2009	[142]
21	PTR-MS	-	Direct	2009	[143]
22	EESI-MS	TATP, TNT, RDX and other explosives were sampled noninvasively from human skin surface	Direct	2009	[144]
23	Electrochemical sensor	Sensitive to TATP vapors	Direct	2009	[145]
24	GC-MPI-MS	-	Direct	2009	[146]
25	GC-IRMS	Method can differentiate the source of the explosive on trace amount of residues	Direct	2009	[147]
26	Electronic sensors array	Sensitive to TATP vapors and not affected by hydrogen peroxide presence	Direct	2009	[148]
27	Colorimetry	Decomposition using 4 M HCl	Indirect (H_2O_2)	2010	[149]
28	Colorimetric sensor array	-	Indirect (H_2O_2, acetone)	2010	[150]
29	Electrochemical sensor	Selective to TATP when compared with HMTD and H_2O_2	Indirect (acetone)	2010	[151]
30	Quartz crystal microbalance	-	Direct	2010	[152]
31	EESI-MS	Coupled to a desorption device that works on different types of surfaces	Direct	2010	[25]
32	IMS-API-MS, portable IMS	-	Direct	2011	[153]
33	NMR, HPLC-UV, HPLC-APCI-MS, IMS	UV detection at 210 nm, ammonium acetate was added and the $[M + NH_4]^+$ ion detected on LC-MS	Direct	2011	[49]
34	ELISA immunoassay	-	Direct	2011	[154]
35	LC-NMR	-	Direct	2011	[59]
36	Gelation	-	Indirect (H_2O_2)	2012	[155]
37	APCI-MS	High-throughput device (1200 persons/hour) installed on a portal	Direct	2012	[156]
38	Chemiluminescence	Selective in relation to HMTD	Direct and indirect (H_2O_2)	2013	[157]
39	Chemiluminescence	Hydrolysis of TATP collected from objects using soaked swabs	Indirect (H_2O_2)	2013	[158]
40	ELISA immunoassay	-	Direct	2013	[159]
41	Colorimetry	A smartphone camera was used as detector and five explosives were differentiated using chemometrics	Direct	2014	[160]
42	GC-MS	On vapor generated from TATP headspace	Direct	2014	[161]
43	Electrochemical sensor	Sensor was molecularly imprinted	Direct	2014	[162]
44	Electrochemical sensor	Sensitive to TATP vapors	Direct	2014	[120]
45	GC-LI-MS	Also applied to HMTD, TNT, RDX and PETN	Direct	2015	[163]
46	Spectrophotometry	TATP was decomposed into hydrogen peroxide using HCl; detection at 554 nm	Indirect (H_2O_2)	2015	[164]
47	GC-MPI-MS	Laser ionization allowed for an increased signal of the molecular ion	Direct	2015	[165]
48	Spectrofluorimetry	-	Indirect (H_2O_2)	2015	[166]
49	Colorimetric sensor array	Also detects HMTD and traces from the synthesis reactants	Direct	2015	[63]
50	EESI-MS	Detection of the $[TATP + Li]^+$ adduct	Direct	2015	[167]
51	DAPP-IMS	On-site TATP and HMTD detection in complex matrices	Direct	2016	[168]
52	GC-IRMS	Samples can be differentiated based on the synthesis reagents' sources	Direct	2016	[169]
53	Electrochemical sensors array	Also sensitive to RDX, TNT and other explosives	Direct	2016	[170]
54	DBDI-MS	TATP and DADP detected without sample preparation steps	Direct	2017	[121]
55	DART-MS	Also applied to other explosives such as HMTD, PETN and RDX.	Direct	2017	[171]
56	Colorimetric and fluorescent sensors	Sensitive to TATP vapor in a closed container	Direct	2017	[172]
57	EASI-MS	Analysis of TATP and DADP on banknote surfaces	Direct	2017	[173]
58	Colorimetry	TATP was decomposed using Amberlyst-15 resin	Indirect (H_2O_2)	2017	[123]
59	Colorimetry	Reaction with radicals generated by thermal decomposition	Indirect (radical species)	2018	[174]
60	FTIR	Also sensitive to HMTD	Direct	2018	[175]
61	HPLC-APCI- MS	Also capable of detecting HMTD	Direct	2018	[176]
62	Colorimetry	-	Direct	2018	[177]
63	Fluorescent sensor	On vapor traces in air	Indirect (H_2O_2)	2019	[178]
64	Colorimetry	Decomposition into hydrogen peroxide using HCl	Indirect (H_2O_2)	2019	[179]

(*continued on next page*)

Table 2 (*continued*)

Entry	Detection method	Observation	Detection	Year	Ref.
65	Electrochemical sensor	Vapor trace detection of TATP and HMTD	Direct	2019	[180]
66	Fluorescent visual sensor	Sensitive to TATP decomposition products vapor	Indirect (acetone)	2019	[181]
67	Fluorescent visual sensor	Sensitive to TATP dissolved in ethanol	Indirect (H_2O_2)	2019	[182]
68	DART-MS	Applied on IED fragments; also capable of detecting HMTD	Direct	2019	[183]
69	FT-ICR-MS	Method also indicates acid used in the synthesis	Direct	2019	[184]
70	PTR-MS	-	Direct	2019	[185]
71	Fluorescent sensor	Also capable of detecting DADP	Indirect (acetone)	2019	[186]
72	Fluorescent sensor	TATP vapor was decomposed using Amberlyst-15 resin	Indirect (H_2O_2)	2019	[187]
73	Colorimetry	-	Direct	2020	[188]
74	Colorimetry, spectrophotometry	Decomposition of TATP by MnO_2 nanoparticles	Indirect (H_2O_2)	2020	[189]
75	Dual-energy X-ray imaging	Using a commercial explosives detection system	Direct	2020	[190]
76	Photoelectric sensor	-	Direct	2020	[191]
77	Raman spectroscopy	Samples analyzed directly on fingerprint and office tapes	Direct	2021	[192]
78	Colorimetry	Thermal decomposition performed in 1 M HCl solution at 60 °C	Indirect (H_2O_2)	2021	[193]
79	Rotational spectroscopy	-	Direct	2021	[194]
80	Fluorescent sensor	Sensitive to TATP vapor in a closed container	Direct	2021	[195]
81	DART-MS	TATP was detected as far as 7.6 m from the DART source	Direct	2021	[196]
82	Fluorescent sensor	Used within a portable microfluidic device	Direct	2021	[197]
83	IMS	Also sensitive to HMTD	Direct	2021	[198]
84	Fluorescent sensor	Tested on environmental samples	Direct	2021	[199]
85	Electrochemical sensor	Increased selectivity due to the use of a molecular imprinted sensor	Direct	2022	[200]
86	Colorimetry	-	Direct	2022	[201]
87	HPLC-ESI-MS	-	Direct	2022	[202]
88	GC-MS	Sample introduction by solid phase microextraction	Direct	2022	[203]
89	Colorimetry	Also capable of detecting HMTD; Decomposition using HCl and H_2SO_4	Indirect (H_2O_2)	2022	[204]
90	Fluorescent sensor	-	Direct	2022	[205]
91	Fluorescent sensor	On vapor traces in air	Direct	2023	[206]
92	HiKE-IMS	Combination of the ammonia-doped and fragmentation ions reduced the probability of false-positives	Direct	2023	[207]
93	GC-MS	Samples collected in passive samplers	Direct	2023	[208]
94	GC-MS	Samples were preconcentrated in metal-organic frameworks-based SPME fibers	Direct	2023	[209]
95	Fluorescent sensor array	Sensitive to TATP vapors	Direct	2023	[210]
96	IMS	Also sensitive to HMTD and other explosives; installed on a gate	Direct	2023	[211]
97	GC-IMS	Simultaneous analysis of 18 explosives including TATP, RDX, PETN and others.	Direct	2024	[212]
98	Fluorescent sensor	Biosensor made of an activated-protein protected gold nanocluster	Direct	2024	[213]
99	LP-acGD-MS	Method was validated for hydrogen peroxide but not for TATP	Direct	2024	[214]
100	IMS	Samples were introduced by headspace solid phase microextraction	Direct	2024	[215]
101	Spectrophotometry	TATP was decomposed through reaction with hydroxylamine hydrochloride	Indirect (H_2O_2)	2024	[216]
102	Electronic sensors array	-	Direct	2024	[217]
103	Spectrofluorimetry and colorimetry	Colorimetric measurements were performed using a smartphone	Direct	2025	[218]
104	GC-MS	Both liquid and headspace samples were analyzed	Direct	2025	[219]
105	GC-IRMS	Method can differentiate the source of the explosive on trace amount of residues	Direct	2025	[220]
106	HPLC-ESI-MS	Also applied on HMTD, DADP and more than 10 other explosives standards	Direct	2025	[221]
107	GC-MS	Using pulsed split injection for superior analytical signal; also applied to several other organic explosives	Direct	2025	[222]

their direct deployment in field settings [121]. Field-relevant detection therefore frequently relies on indirect strategies, such as conversion of **TATP** to hydrogen peroxide followed by colorimetric or optical readout, exploiting the higher chemical reactivity and detectability of peroxide species [124].

Colorimetric field-oriented strategies for the detection/identification of peroxide-based explosives often rely on converting **TATP** into a more readily detectable analyte, most commonly H_2O_2, followed by a chromogenic readout. One approach explicitly hydrolyzes **TATP** to H_2O_2 using a strongly acidic cation-exchanger (Amberlyst-15) and then quantifies the generated H_2O_2 via a nanoparticle-driven colorimetric scheme, reporting LODs of 20 nM for H_2O_2 and 0.31 mg L^{-1} for **TATP** [123]. This indirect approach is driven by practical forensic limitations, including the absence of spectroscopically active functionalities, such as nitro groups or aromatic systems, in peroxide explosives. In addition, colorimetric detection of **TATP** is significantly simplified after its hydrolysis to H_2O_2. Screening is further hindered by the volatility and residue characteristics of **TATP**, as it readily sublimes at room temperature and, unlike typical nitro-explosives, does not produce post-blast residues [123].

Environmental and aqueous detection introduces additional constraints. Reviews addressing explosives in water highlight that residues of **TATP** and related precursors may enter wastewater streams, yet matrix effects, low concentrations, and rapid transformation limit direct detection [7]. Sample-preparation steps such as solid-phase extraction are therefore required to achieve meaningful detection limits, increasing analytical complexity and time. Vapor-phase detection faces parallel challenges, as low equilibrium vapor pressures and variability linked to synthesis conditions complicate quantitative interpretation and sensor calibration [87].

Taken together, current analytical strategies for **TATP** rely on a combination of direct molecular identification, indirect peroxide sensing, and matrix-specific workflows, each with inherent limitations. No single method simultaneously satisfies the requirements of sensitivity, selectivity, portability, and robustness across all scenarios. In this context, Table 2 provides a consolidated overview of the principal analytical methods reported for the detection and quantification of **TATP**, summarizing their operating principles, detectable species, analytical performance, and practical constraints, thereby offering a comparative framework for method selection and future development.

7. Concluding remarks and trends

The body of work consolidated in this review demonstrates that **TATP** occupies a unique position among energetic materials, defined less by energetic density and more by extreme kinetic and structural fragility. Across molecular, crystalline, and macroscopic scales, its behavior is governed by weak O–O σ bonds, minimal intermolecular

stabilization, and entropy-dominated decomposition pathways. These features explain why **TATP** behaves as a primary explosive despite its low density and absence of nitro functionalities, and why small perturbations in phase, environment, or handling conditions can shift its response from benign degradation to violent detonation.

A central theme emerging from recent studies is the critical role of solid-state and gas-phase dynamics in shaping hazard profiles. Polymorphism, solid-solid transformations to **DADP**, and continuous sublimation establish that **TATP** samples are intrinsically dynamic systems rather than static solids. These processes directly influence sensitivity, vapor exposure, and analytical detectability, reinforcing the need to consider time, storage conditions, and confinement when assessing risk. The convergence of crystallographic, thermochemical, and gas-phase investigations has clarified that detonation initiation cannot be attributed to a single trigger but arises from coupled structural and physicochemical factors.

Progress in deactivation and mitigation strategies highlights both opportunities and limitations. Chemical neutralization via redox-active species, aggressive oxidants, or solution-phase treatments is feasible, yet operates within a narrow safety window where heat and gas evolution must be carefully managed. Materials-assisted approaches, including porous frameworks, molecularly imprinted polymers, and metal oxides, illustrate a shift toward passive mitigation and interception strategies that rely on confinement, selective binding, or surface-mediated interactions rather than bulk destruction. These approaches are particularly relevant for detection and prevention, where inducing detonation must be avoided at all costs.

From an analytical perspective, no universal method for **TATP** detection has yet emerged. Instead, current practice relies on complementary techniques that target intact molecules, degradation products, or indirect peroxide signatures, each with inherent trade-offs between sensitivity, selectivity, portability, and robustness. The growing integration of analytical chemistry with mechanistic understanding suggests a trend toward hybrid strategies that exploit controlled chemical interactions for both sensing and mitigation.

Looking forward, advances in materials design, coupled with refined models of initiation and decomposition, are expected to improve risk assessment frameworks and countermeasure development. Ultimately, a chemically informed, system-level approach remains essential for addressing the persistent challenges posed by **TATP** in forensic, safety, and security contexts.

CRediT authorship contribution statement

Brenno A.D. Neto: Conceptualization, Formal analysis, Funding acquisition, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Gilson L.C.L. Filho:** Conceptualization, Writing – original draft. **Alan R. Mol:** Conceptualization, Writing – original draft. **Alexandre Fonseca:** Conceptualization, Writing – original draft. **Valter H. Carvalho-Silva:** Conceptualization, Writing – original draft. **Peter Bakuzis:** Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No primary research results and no new data were generated or analyzed as part of this review.

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